

Assignment: Cloud Computing

Module: 1

1) What is Cloud Computing?

Answer:

- **Cloud computing is the delivery of computing services like servers, storage, databases, networking, and software over the internet.**
 - **It allows users to access resources on demand without managing physical hardware.**
 - **Cloud services are scalable, flexible, and cost-effective.**
- Examples include AWS, Microsoft Azure, and Google Cloud.**

2) Describe Cloud Computing Deployment Models

Answer:

1. Public Cloud

- **Services are provided over the internet for public use.**
- **Example: AWS, Azure.**

2. Private Cloud

- **Cloud infrastructure is dedicated to a single organization.**
- **Provides better security and control.**

3. Hybrid Cloud

- **Combination of public and private cloud environments.**
- **Allows data and application sharing.**

4. Community Cloud

- **Shared by multiple organizations with common requirements.**

3) What are Components of Cloud Computing?

Answer:

Main components of cloud computing are:

- **Frontend – User devices and applications.**
- **Backend – Servers, databases, and cloud infrastructure.**
- **Cloud Storage – Stores data online.**
- **Network – Internet connection for communication.**
- **Virtualization – Creates virtual machines and resources.**

4) Advantages and Disadvantages of Cloud Computing

Advantages:

- **Cost-effective**
- **Scalable resources**
- **Remote access from anywhere**
- **Automatic backup and updates**
- **High availability**

Disadvantages:

- **Internet dependency**
- **Security and privacy concerns**
- **Downtime issues**
- **Limited control over infrastructure**

Module: 2

1) What is Virtualization and its Types?

Answer:

Virtualization is the process of creating virtual versions of servers, storage, or operating systems.

It allows multiple virtual machines to run on a single physical machine.

Types:

- **Server Virtualization**
- **Storage Virtualization**
- **Network Virtualization**
- **Desktop Virtualization**

2) Types of Hypervisor and How to Manage It

Answer:

Type 1 Hypervisor

- **Runs directly on hardware.**
- **Example: VMware ESXi, Hyper-V.**

Type 2 Hypervisor

- **Runs on top of an operating system.**
- **Example: VirtualBox, VMware Workstation.**

Management:

- **Managed using web consoles or management software.**
- **Used to create, monitor, and manage virtual machines.**

3) Roles of Virtualization in Cloud Computing

Answer:

- **Improves hardware utilization**
- **Reduces infrastructure cost**
- **Enables resource sharing**
- **Supports scalability and flexibility**
- **Simplifies backup and disaster recovery**

4) What is Container?

- **Answer:**
A container is a lightweight package containing application code and dependencies.
 - **Containers run consistently across different environments.**
Docker and Kubernetes are commonly used container technologies.
 - **Containers are faster and use fewer resources than virtual machines.**

5) What is High Availability and Live Migration in Virtualization?

Answer:

High Availability (HA)

- **Ensures systems remain operational during failures.**
- **Minimizes downtime.**

Live Migration

- **Transfers a running virtual machine from one host to another without shutdown.**
- **Used for maintenance and load balancing.**

6) Describe Block Storage, File Storage, Object Storage, DAS, NAS, and SAN

Answer:

Block Storage

- **Data stored in fixed-size blocks.**
- **Used in databases and virtual machines.**

File Storage

- **Data stored as files and folders.**
- **Shared over networks.**

Object Storage

- **Stores data as objects with metadata.**
- **Used for cloud backups and multimedia.**

DAS (Direct Attached Storage)

- **Storage directly connected to a computer.**

NAS (Network Attached Storage)

- **Shared storage connected over a network.**

SAN (Storage Area Network)

- **High-speed network providing block-level storage.**

7) Describe Storage Allocation and Provisioning

Answer:

- **Storage allocation means assigning storage space to users or applications.**
 - **Provisioning is the process of creating and configuring storage resources.**
 - **It helps optimize storage usage and improve performance.**
- **Cloud platforms provide automatic storage provisioning features.**

Module: 3

1) Different Types of Cloud Storage

Answer:

- **Object Storage**
- **File Storage**
- **Block Storage**
- **Archive Storage**
- **Backup Storage**

Each type is used for different cloud storage requirements.

2) What is RBAC, IAM, and MFA?

Answer:

RBAC (Role-Based Access Control)

- **Provides permissions based on user roles.**

IAM (Identity and Access Management)

- **Manages users and access permissions.**

MFA (Multi-Factor Authentication)

- **Adds extra security using OTP, biometrics, or authentication apps.**

3) What is Physical and Virtual Host Allocation?

Answer:

Physical Host Allocation

- **Assigning physical hardware resources to systems.**

Virtual Host Allocation

- **Assigning virtual resources using virtualization technology.**

Virtual allocation improves flexibility and resource utilization.

4) How to Access Resources of Cloud Computing?

Answer:

Cloud resources are accessed using:

- **Web portals**
- **APIs**
- **Remote desktop tools**
- **SSH and RDP protocols**
- **Cloud management software**

5) Types of Backup in Cloud

Answer:

- **Full Backup**
- **Incremental Backup**
- **Differential Backup**
- **Snapshot Backup**
- **Remote Backup**

Backups help recover data during failures.

6) What is Disaster Recovery?

Answer:

- **Disaster Recovery (DR) is the process of restoring systems and data after failures or disasters.**
- **It ensures business continuity and minimizes downtime.**
- **Cloud platforms provide automated DR solutions and backups.**

Module: 4

1) Resource Monitoring Techniques

Answer:

Resource monitoring tracks CPU, memory, storage, and network usage. Tools used include CloudWatch, Azure Monitor, and Prometheus. Monitoring helps improve performance and detect issues early.

2) How to Access Compute (Windows and Linux) from Internet?

Answer:

Windows Access

- **Uses RDP (Remote Desktop Protocol).**

Linux Access

- **Uses SSH (Secure Shell).**

Security:

- **Use firewalls**
- **Enable MFA**
- **Use strong passwords**
- **Restrict IP access**

3) Encryption Technologies and Methods

Answer:

Encryption converts data into unreadable form for security.

Types:

- **Symmetric Encryption**
- **Asymmetric Encryption**

- **AES Encryption**
- **RSA Encryption**

4) Describe Network Security, Compute Security, and Storage Security

Answer:

Network Security

- **Protects networks using firewalls, VPNs, and IDS.**

Compute Security

- **Secures virtual machines and servers.**

Storage Security

- **Protects stored data using encryption and backups.**

Module: 5

1) How to Configure, Develop and Maintain Security and Privacy in Cloud?

Answer:

- **Use IAM and RBAC**
- **Enable encryption**
- **Configure firewalls**
- **Apply regular updates and patches**
- **Monitor logs and activities**
- **Enable backups and MFA**

2) What is Portability in Cloud?

Answer:

Portability means moving applications and data between different cloud platforms without major changes.

It helps avoid vendor lock-in and improves flexibility.

3) What is Reliability and High Availability in Cloud?

Answer:

Reliability ensures cloud services work consistently without failure.

High Availability ensures systems remain accessible with minimal downtime.

Cloud providers use redundant servers and backups for HA.

4) Describe Mobility Cloud Computing

Answer:

Mobility cloud computing allows users to access cloud services using

mobile devices from anywhere.

It supports remote work, synchronization, and real-time access to applications.

5) Describe AWS, Azure, and Google Cloud Platforms

Answer:

AWS

- Cloud platform by Amazon.
- Provides compute, storage, networking, and AI services.

Azure

- Microsoft cloud platform.
- Supports Windows and enterprise applications.

Google Cloud Platform (GCP)

- Google's cloud platform.
- Known for AI, analytics, and Kubernetes support.

6) Accessing AWS, Azure, and Google Cloud Platforms

Answer:

Users can access cloud platforms through web portals:

- AWS Management Console
- Azure Portal
- Google Cloud Console

Users log in using credentials and manage cloud resources online.

7) Create Compute, Network, and Storage on AWS, Azure, and GCP

Answer:

Compute

- **AWS EC2**
- **Azure Virtual Machine**
- **Google Compute Engine**

Network

- **VPC in AWS**
- **Virtual Network in Azure**
- **VPC in GCP**

Storage

- **AWS S3**
- **Azure Blob Storage**
- **Google Cloud Storage**

8) Compare Cloud Pricing of AWS, Azure, and GCP

Answer:

AWS

- **Flexible pay-as-you-go pricing.**
- **Large service availability.**

Azure

- **Cost-effective for Microsoft environments.**
- **Good enterprise integration.**

GCP

- **Competitive pricing with discounts.**
- **Strong in AI and analytics services.**